

Physics-Informed Neural Networks for PDEs

How to build PDE structure into neural solvers

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Motivation

Why use neural networks for PDEs?

They provide flexible function approximators that can be evaluated rapidly after training and can represent high-dimensional solution maps.

Why early approaches are limited

Purely data-driven neural PDE solvers typically:

- require numerical or sampled solution data,
- inherit the cost and bias of that data source,
- may fit values well while violating the underlying PDE structure,
- do not automatically respect boundary conditions, admissibility, or variational structure.

Main idea of this project

Start from a standard uninformed PINN and ask:

How can PDE theory be encoded more directly into the model?

The answer depends on the equation class:

- parabolic: traces and comparison structure,
- hyperbolic: weak solutions and entropy,
- elliptic: admissible spaces and variational structure.

Goal

Enhance a naive neural solver by building in mathematically meaningful constraints rather than treating the PDE as just another regression target.



Deep Learning Background

General Intro to Deep Learning

Neural ansatz

A feedforward network is a parameterized map

$$z \mapsto u_{\theta}(z),$$

where z is the input coordinate and θ is the collection of weights and biases.

Why this is useful for PDEs

If u_{θ} is differentiable, then automatic differentiation gives

$$u_{\theta,x}, \quad u_{\theta,t}, \quad u_{\theta,xx}, \quad \dots$$

so the network can be inserted directly into a PDE, boundary condition, or energy functional.

Loss functions

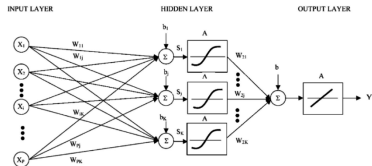
Training is parameter optimization:

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\theta).$$

In ordinary ML, $\mathcal{L}$ measures mismatch with labels.

In PDE problems, $\mathcal{L}$ may contain:

- residual loss,
- boundary / initial loss,
- variational energy,
- entropy or barrier penalties.



Uninformed PINNs

Baseline formulation

For a PDE

$$\mathcal{N}[u] = f,$$

the standard PINN uses a neural ansatz u_θ and minimizes a strong residual:

$$\mathcal{L}_{\text{res}} = \frac{1}{N} \sum_{j=1}^N |\mathcal{N}[u_\theta](z_j) - f(z_j)|^2.$$

Boundary and initial conditions are then usually added as soft penalties:

$$\mathcal{L} = \mathcal{L}_{\text{res}} + \lambda_b \mathcal{L}_{\text{BC}} + \lambda_i \mathcal{L}_{\text{IC}}.$$

Advantage

Very general: the same template can be used for many PDEs.

Main weakness

The model is only asked to reduce sampled losses. It does not naturally know about admissible spaces, weak formulations, entropy conditions, or maximum principles.



Hard and Soft Constraint Enforcement

Soft enforcement

A constraint is imposed through a penalty term.

Example:

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_b} \sum_{k=1}^{N_b} |u_{\theta}(\xi_k) - g(\xi_k)|^2.$$

The optimizer must trade off PDE residual against constraint mismatch.

Hard enforcement

A constraint is built into the trial space itself:

$$u_{\theta} = g^* + \phi N_{\theta},$$

where $\phi = 0$ on the relevant boundary or initial set.
Then every admissible network output satisfies the constraint identically.

Why this matters

These are not just implementation choices. They change what class of functions the optimizer is searching over, and therefore change the mathematical meaning of the learned approximation.



Parabolic Problems

Parabolic Benchmark Problem

Equation

$$u_t - u_{xx} + 2u_x = f(x, t), \quad (x, t) \in [0, 1] \times [0, 1].$$

Exact solution:

$$u(x, t) = e^{-t} \sin(\pi x) + t x(1 - x).$$

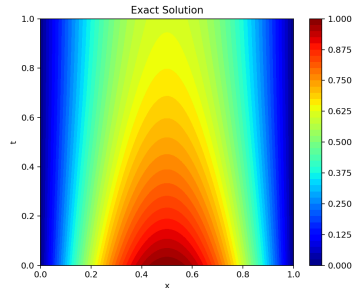
Initial and boundary data:

$$u(x, 0) = \sin(\pi x), \quad u(0, t) = u(1, t) = 0.$$

Why this benchmark works

It has:

- exact solution,
- nontrivial space-time evolution,
- separate roles for interior, initial, and boundary data.



Parabolic Constraint: Hard Initial and Boundary Enforcement

Trial-space construction

Use the ansatz

$$u_{\theta}(x, t) = \sin(\pi x) + t x(1 - x) N_{\theta}(x, t).$$

Then:

$$u_{\theta}(x, 0) = \sin(\pi x), \quad u_{\theta}(0, t) = u_{\theta}(1, t) = 0.$$

Mathematical meaning

This is not just a numerical trick. The approximation is forced into the correct trace class from the start.

Effect

The optimizer no longer has to trade off the PDE residual against initial/boundary mismatch.



Parabolic Constraint: Comparison Principle and Barriers

Comparison principle

For a parabolic operator, if

$$\mathcal{L}\underline{u} \leq \mathcal{L}\bar{u}$$

in the domain and

$$\underline{u} \leq \bar{u}$$

on the parabolic boundary, then one expects the ordering

$$\underline{u} \leq u \leq \bar{u}$$

to persist in the interior.

Chosen barriers

$$\underline{u}(x, t) = e^{-t} \sin(\pi x),$$

$$\bar{u}(x, t) = e^{-t} \sin(\pi x) + 2tx(1-x).$$

Loss term

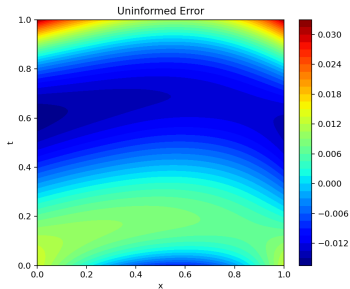
Penalize violations of

$$\underline{u} \leq u_\theta \leq \bar{u}.$$

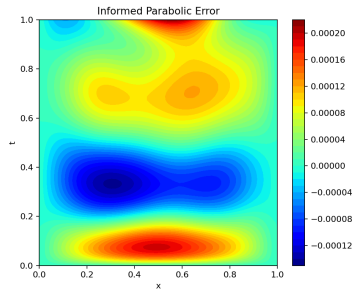
This softly encodes qualitative parabolic structure beyond the residual.



Parabolic Results



Uninformed: broad signed-error pattern over the whole space-time domain



Informed: much smaller absolute error, localized in narrow bands

Key metrics

$$\text{rel } L^2 \approx 1.39 \times 10^{-4}, \quad \text{heldout residual} \approx 7.62 \times 10^{-6},$$

$$\text{final-time rel } L^2 \approx 2.50 \times 10^{-4}, \quad \text{mean barrier violation} \approx 5.85 \times 10^{-10}.$$



Hyperbolic Problems

Hyperbolic Benchmark Problem

Burgers shock problem

$$u_t + \left(\frac{u^2}{2} \right)_x = 0.$$

Initial data:

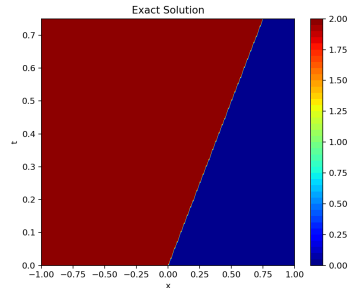
$$u(x, 0) = 2 \text{ for } x < 0, \quad u(x, 0) = 0 \text{ for } x > 0.$$

Rankine-Hugoniot speed:

$$s = \frac{f(2) - f(0)}{2 - 0} = 1.$$

Entropy solution:

$$u(x, t) = 2 \text{ for } x < t, \quad u(x, t) = 0 \text{ for } x > t.$$



Hyperbolic Constraint: Weak Solutions Are Not Enough

Weak formulation

For a scalar conservation law

$$u_t + f(u)_x = 0,$$

classical solutions may break down after characteristics intersect. One therefore enlarges the solution class and asks that the PDE hold in the weak sense.

Main issue

Weak solutions need not be unique after shock formation.

So a neural network can reduce a sampled strong residual and still fail to select the physically relevant shock.

For hyperbolic PDEs, the problem is not just enforcing the equation — it is selecting the right weak solution.



Hyperbolic Constraint: Entropy Admissibility

Entropy inequality

For each convex entropy η with entropy flux q , the admissible solution satisfies

$$\eta(u)_t + q(u)_x \leq 0$$

in the weak sense.

For Burgers

$$\eta(u) = \frac{u^2}{2}, \quad q(u) = \frac{u^3}{3}.$$

Loss term

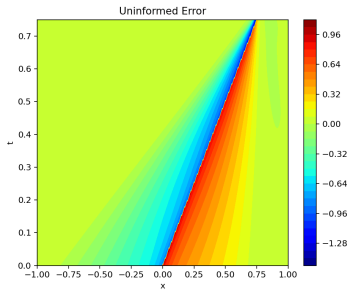
Penalize the positive part of

$$\eta(u_\theta)_t + q(u_\theta)_x.$$

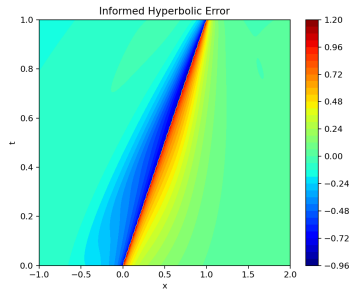
This drives the model toward the entropy solution class.



Hyperbolic Results



Uninformed: broad wedge-shaped error around the shock



Informed: error confined near the shock trajectory

Key metrics

$$\begin{aligned} \text{rel } L^1 &\approx 1.57 \times 10^{-1}, & \text{final-time rel } L^1 &\approx 7.04 \times 10^{-2}, \\ \text{shock location error} &\approx 2.5 \times 10^{-3}, & \text{undershoot} &\approx 5.08 \times 10^{-3}. \end{aligned}$$

The informed model does not remove all error — it organizes the error around the correct entropy shock.



Elliptic Problems

Elliptic Benchmark Problem

Variable-coefficient Dirichlet problem

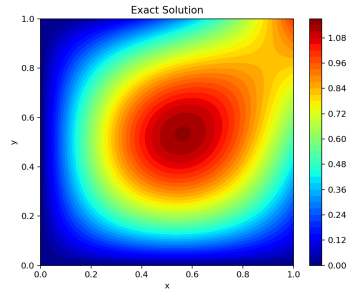
$$-\nabla \cdot (a(x, y) \nabla u) + c(x, y)u = f(x, y), \quad \Omega = (0, 1)^2.$$

Exact solution:

$$u(x, y) = \sin(\pi x) \sin(\pi y) + x^2 y.$$

Coefficients:

$$a(x, y) = 1 + x + y, \quad c(x, y) = 1 + x^2.$$



Elliptic Constraint: Dirichlet Traces and Admissible Classes

Dirichlet viewpoint

For elliptic boundary-value problems, the boundary condition is part of the admissible class. In weak theory one solves in a space such as

$$H_g^1(\Omega) \quad \text{or} \quad H_0^1(\Omega),$$

not in an unrestricted function class.

Neural ansatz

Use

$$u_\theta(x, y) = g^*(x, y) + x(1-x)y(1-y) N_\theta(x, y).$$

Because $x(1-x)y(1-y) = 0$ on $\partial\Omega$, the neural correction vanishes on the boundary and the trace is exact.



Elliptic Constraint: Ritz Formulation and Euler–Lagrange Structure

Variational principle

For a coercive elliptic PDE, the solution can often be characterized as the minimizer of an energy functional

$$\mathcal{J}(v) = \int_{(0,1)^2} \left[\frac{1}{2}(1+x+y)|\nabla v|^2 + \frac{1}{2}(1+x^2)v^2 - f(x,y)v \right] dx dy$$

The Euler–Lagrange equation for $\mathcal{J}$ is the PDE itself.

Baseline

Strong residual collocation:

$$\mathcal{L}_{\text{res}}.$$

Informed model

Ritz training:

$$u_\theta \approx \arg \min \mathcal{J}(u_\theta).$$

This ties the neural solver directly to Sobolev-space and Euler–Lagrange theory rather than only to pointwise residual minimization.



Elliptic Constraint: Maximum/Comparison Principle

Principle

For

$$Lu = -\nabla \cdot (a \nabla u) + cu, \quad c \geq 0,$$

the elliptic comparison principle says boundary data control interior extrema. In particular, if

$$Lu \geq 0 \text{ in } \Omega, \quad u \geq 0 \text{ on } \partial\Omega,$$

then

$$u \geq 0 \text{ in } \Omega.$$

Benchmark operator

$$a(x, y) = 1 + x + y > 0, \quad c(x, y) = 1 + x^2 \geq 1,$$

so this problem lies in the standard elliptic max/min-principle setting.

Benchmark bounds

For

$$u(x, y) = \sin(\pi x) \sin(\pi y) + x^2 y,$$

we have

$$0 \leq \sin(\pi x) \sin(\pi y) \leq 1, \quad 0 \leq x^2 y \leq 1,$$

hence

$$0 \leq u(x, y) \leq 2.$$

What was enforced

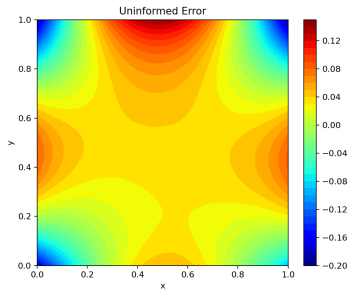
The qualitative loss penalized violations of

$$0 \leq u_\theta(x, y) \leq 2.$$

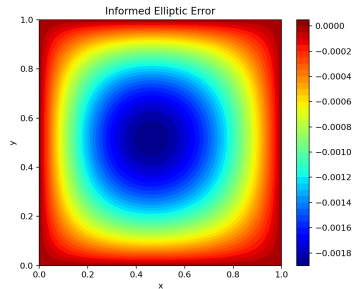
So the lower bound 0 reflects the maximum/comparison-principle viewpoint, while 2 is a simple analytic upper bound for this benchmark.



Elliptic Results



Uninformed: broad signed error, strongest near the boundary and corners



Informed: small interior absolute error, essentially zero on the boundary

Key metrics

$$\text{rel } L^2 \approx 1.51 \times 10^{-3}, \quad \text{rel } H^1\text{-semi} \approx 2.07 \times 10^{-3},$$

$$\text{boundary error} \approx 1.23 \times 10^{-7}, \quad \text{mean barrier violation} = 0.$$



Comparisons

Comparisons and Repeated-Seed Averages

PDE	Solution metric	Structure-aware metric 1	Structure-aware metric 2
Parabolic	$\text{rel } L^2 = (2.70 \pm 1.30) \times 10^{-4}$	$\text{final-time rel } L^2 = (2.73 \pm 1.06) \times 10^{-4}$	mean barrier violation $= 5.85 \times 10^{-10}$
Hyperbolic	$\text{rel } L^1 = 0.166 \pm 0.019$	shock location error $= 0.015 \pm 0.015$	mean entropy violation $= 0.0589 \pm 0.0080$
Elliptic	$\text{rel } L^2 = (1.43 \pm 0.63) \times 10^{-3}$	$\text{rel } H^1\text{-semi} = (2.16 \pm 1.03) \times 10^{-3}$	boundary trace error $= 1.23 \times 10^{-7}$

Interpretation

The informed models consistently improve on the uninformed baseline, but the relevant gain depends on the PDE: trace accuracy for parabolic, admissibility for hyperbolic, and variational/boundary structure for elliptic.



Extensions

Extensions

Constraint and loss design

- monotonicity, convexity, conservation, positivity,
- augmented-Lagrangian or Lagrange-multiplier enforcement,
- adaptive collocation near boundaries and shocks,
- sparse data augmentation and hybrid physics-plus-data training.

Uncertainty

- repeated-seed variability,
- sampling-induced uncertainty,
- Bayesian PINNs.

Broader formulations

- first-order PINNs for conservation laws,
- weak-form and domain-decomposed PINNs,
- more general coefficients and geometries,
- inverse problems.

Architecture

- neural operators,
- DeepONet / Fourier neural operator,
- structure-aware architectures such as Lagrangian or Hamiltonian networks.



Thank you

Questions?

github.com/zane-perry/pde-pinns

